

# Public engagement with health and climate change around the world: a Google Trends analysis

Niheer Dasandi, Slava Jankin, Dafni Kalatzi Pantera, Marina Romanello



Despite growing recognition of the importance of people engaging with the health dimensions of climate change, we know surprisingly little about the levels of public engagement around the world. We address this knowledge gap by examining Google Trends data, using people's online information-seeking behaviour to shed light on global engagement with health and climate change between 2014 and 2023. We observe that over the past decade—and particularly since 2020—there has been growing public engagement via Google searches with health and climate change around the world. The increasing engagement with the intersection of health and climate change is largely distinct from engagement with either climate change or health separately. We observe that such engagement is highest in low-income and middle-income countries. There is also greater engagement with health and climate change than with other issues that intersect climate change—eg, the economy and security—highlighting the public salience of health framings of climate change.

## Introduction

Over the last decade, there has been growing attention to how people understand and engage with the health dimensions of climate change. Many people view public awareness of the relationship between health and climate change as fundamental to strengthening support for climate action and increasing an individual's willingness to adopt more pro-environmental behaviours.<sup>1–4</sup> This push to recognise health and climate change together is based on the argument that a health framing can shift public attitudes by making climate change more personally relevant, immediate, and geographically proximate—and can further motivate climate action by emphasising the positive health co-benefits of such action rather than focusing solely on the negative consequences of inaction.<sup>1,4,5</sup> An expanding evidence base supports this view, with studies finding that a health framing can increase public engagement with climate change and strengthen public support for mitigation policies, including in individuals less concerned with the effects of climate change.<sup>3,6–10</sup> Yet, despite these findings, we know surprisingly little about the level of public engagement with health and climate change, how this engagement varies across countries, and how it has changed over time.<sup>11</sup>

This knowledge gap stems from the absence of relevant data about public attitudes to health and climate change that is of sufficient geographical and temporal coverage and resolution. The principal approach to measuring public opinion is by conducting surveys. By asking people to respond to specific questions, such surveys provide invaluable and comparable information about the level (and different types) of public engagement with climate change and health. However, surveys provide information about public attitudes in fixed geographical locations at specific points in time, and unless they are repeated periodically and across countries, they tell us little about global trends in public engagement over time.<sup>12</sup> This challenge is considerable given the high costs of

conducting surveys with representative samples.<sup>13</sup> Consequently, there have been few surveys on public attitudes to health and climate change and the surveys that have been conducted have been one-off surveys in a handful of high-income countries (HICs).<sup>14–17</sup> In an effort to gain a better understanding of the patterns of public engagement with health and climate change, some researchers have turned to data from social media platforms, such as X (formerly Twitter).<sup>18,19</sup> However, barriers to data access and geolocating posts mean that such analyses focus on individual years, a restricted number of countries, and a small sample of posts. The use of X also varies widely across regions and demographic groups.<sup>20</sup> Furthermore, X's decision to discontinue free access to its Academic API in February, 2023 means that these data are largely inaccessible for most researchers going forward.

Such obstacles leave crucial questions about public engagement with health and climate change around the world largely unanswered: has public engagement with health and climate change around the world increased over time? In which regions and countries do we see the highest levels of public engagement with climate change and health? How does engagement with health and climate change compare with public engagement with other climate change issues and impacts? This study aims to address these gaps using an alternative approach to measuring public engagement based on the extent to which people sought information about health and climate change on the Google search engine for the 10-year period between 2014 and 2023. Google is the most visited website in the world with people using the search engine to find information related to all aspects of their lives.<sup>21</sup> Hence, using Google search data allows us to measure a key dimension of public engagement, namely people's online information-seeking behaviour. Furthermore, these data are available over time and across countries, enabling us to examine trends in public engagement around the world.

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School of Government (Prof N Dasandi PhD, D Kalatzi Pantera PhD) and The Centre for Artificial Intelligence in Government

(Prof S Jankin PhD), University of Birmingham, Birmingham, UK; Institute for Global Health, University College London, London, UK (M Romanello PhD)

Correspondence to: Prof Niheer Dasandi, School of Government, University of Birmingham, Birmingham B15 2TT, UK [n.dasandi@bham.ac.uk](mailto:n.dasandi@bham.ac.uk)

## Approach

We examine public engagement with health and climate change using Google Trends data. Public engagement comprises a range of cognitive, affective, and behavioural dimensions—such as levels of knowledge about climate change–health links, concerns about specific health impacts, or adopting personal mitigation behaviours with health co-benefits (eg, reducing red meat consumption).<sup>22</sup> We focus on a single, but substantial, behavioural dimension of engagement: people's online information-seeking behaviour. These data are an especially meaningful indicator of wider public engagement because they are closely connected with various cognitive, affective, and behavioural dimensions of engagement. For example, online information-seeking behaviour can foster a better understanding of the climate change–health relationship (and itself might result from greater awareness of this relationship), which might in turn lead an individual to adopt behavioural actions, such as engaging in advocacy or adopting pro-environmental behaviours.

When examining people's online climate change–health information-seeking behaviour, we focused specifically on use of the Google search engine. As the most visited website in the world, the Google search engine in 2023–24 had a 91% share of the global search engine market, processing over 3·5 billion search queries per day.<sup>21,23</sup> Google Trends provides web-query data that captures average search patterns on Google's search engine among populations by country over time. These data have been used in public health research for more than two decades primarily to monitor and analyse human behaviour using information-seeking behaviour on the Google search engine, and thereby assess, predict, and potentially prevent health-related issues that arise around the world.<sup>24–27</sup> Our use of Google Trends data is similar to such studies in that it seeks to capture individuals' information-seeking behaviour. However, we aim to shed light on the extent of public engagement with health and climate change around the world based on people's information-seeking behaviour using the Google search engine. This approach is in line with political science studies that use Google search data to measure the public salience of issues, such as human rights and global trade.<sup>12,28–30</sup>

See Online for appendix

Our analysis assumes that the extent to which people seek information online about health and climate change using the Google search engine provides an indicator of levels of public engagement with health and climate change around the world. We use Google Trends data for searches of health and climate change to examine trends in public engagement over the 10-year period between Jan 1, 2014 and Dec 31, 2023. This approach helps address gaps in our understanding of public engagement with health and climate change globally. There are, nonetheless, limitations of using Google Trends to measure engagement with health and climate change. For example,

Google is banned in China meaning the country is excluded from our analysis. More generally, internet penetration rates vary considerably around the world, which means Google Trends data might exclude some groups and regions in low-income and middle-income countries (LMICs), affecting the representativeness of data from these countries.<sup>31,32</sup> Furthermore, Google periodically updates its data collection method with the aim of enhancing its algorithms for more accurate identification of user queries; however, this has produced differences in values obtained before and after such updates.<sup>33</sup>

A notable limitation of using Google Trends data to examine engagement is that raw search volume data are not publicly accessible. Instead, Google provides a normalised index of search interest. This index is derived from the volume of searches containing the terms of interest (eg, “climate change” and “health”) relative to the total search volume for a defined population and time period. By normalising against total search volume, the data can be compared over time, accounting for overall changes in Google search engine usage. Google Trends scales the normalised data from 0 to 100, where a value of 100 is assigned to the point of highest popularity for the search term within the specified time range and region. All other datapoints are expressed as a proportion of this peak. For example, a value of 50 indicates that the search term was half as popular at that point compared with the peak. A score of 0 does not necessarily mean no searches, but that the search volume did not meet a minimum threshold for inclusion and hence indicates a very low search rate. Google Trends makes this transformed and scaled data publicly available, which we used to measure public engagement with climate change and health. While this approach has limitations, it provides valuable insights into the relative salience of these issues over time and across different populations.

We used two different approaches to collecting Google Trends data on health and climate change. First, we used “health” and “climate change” together as a search term, whereby we identified Google searches that contain both terms in different languages. We conducted this search in five languages: English, Spanish, French, Portuguese, and Arabic (including combinations of these languages). The appendix (pp 2–6) presents searches conducted using additional languages.

Second, we used an approach based on Google Trends topics and categories. Google Trends provides search ratios for topics in addition to individual search terms, whereby topics are bundles of search terms that share the same concept in any language. In other words, using the topic, “climate change” includes all searches of climate change irrespective of the language in which the search is conducted. Google Trends provides data for the “climate change” topic, but it does not feature a “climate change health” topic. Google also classifies search queries into categories and sub-categories using an

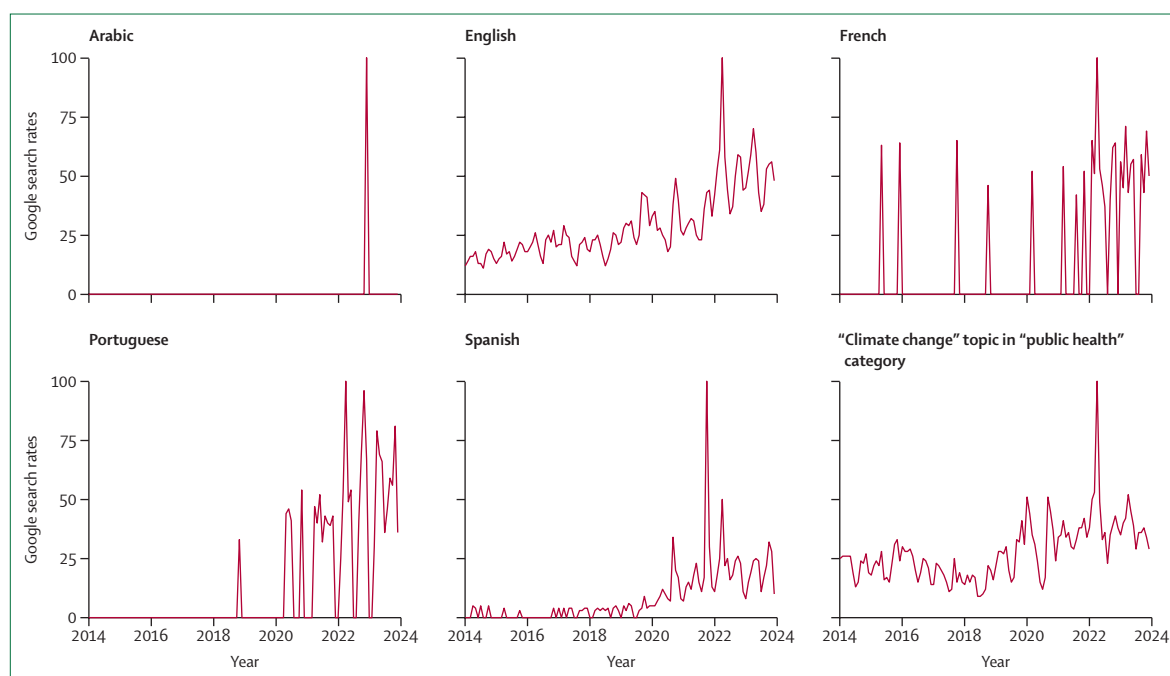


Figure 1: Global monthly search rates of health and climate change from Google Trends for five languages and the topic-category approach, 2014–23

automated classification engine based on natural language processing methods.<sup>34</sup> Categories are broader than topics and effectively connect a search term or topic to the specific context in which they are being investigated. As such, by specifying a category, the Google search query data are restricted to those searches made in the context of the specified category. Accordingly, we collected Google Trends data on the “climate change” topic that have been classified as belonging to the “public health” category. This approach provides search data for “climate change” across all languages, which are specifically related to a public health context. We refer to this second approach as the topic-category approach. Our analysis of public engagement with health and climate change used both approaches. With the topic-category approach being language agnostic, another considerable difference between the two approaches is that the first only includes searches that explicitly mention “health” (in addition to “climate change”); while the second includes searches of “climate change” that might not explicitly mention “health”, but instead reference public health-related issues, such as specific diseases or health services, and therefore might include a larger volume of searches (appendix p 3).

We also draw on other features of the Google Trends data platform. The first is that the Google Trends approach based on normalised transformations of the raw search data also allows for relative comparisons of search interest over time and between different terms.<sup>35</sup> The Google Trends platform provides data on relative comparisons across different search terms for defined populations over a fixed time period,<sup>36</sup> which enabled us

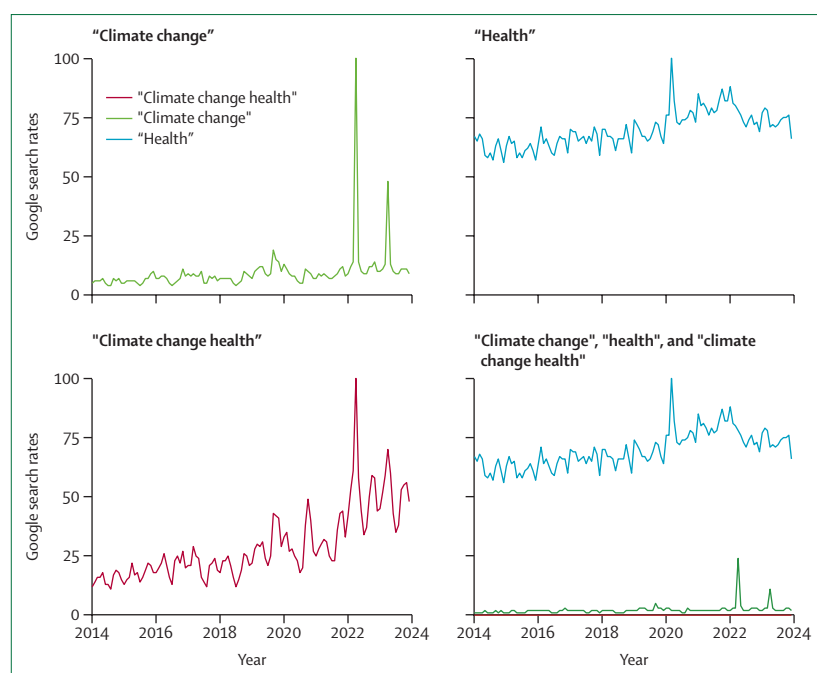
to compare relative search rates of climate change and health with other relevant search terms, such as “climate change impacts” or “climate change security”. Second, Google provides information about queries and topics that are related to the main search terms and topics of interest.<sup>12,37</sup> In other words, by specifying searches for “health” and “climate change” in a given language for the 10-year time period, we can also observe related search terms in that language.

## Insights

What do Google searches around the world reveal about public engagement with climate change and health? We consider several areas to address this question. First, we examined global trends in engagement over a 10-year period (2014–23). Second, we investigated engagement across different countries to assess which regions have the highest levels of public engagement with the health dimensions of climate change. We derived several key insights about public engagement with health and climate change around the world.

## Trends in global engagement over time

We first considered trends in public engagement with health and climate change around the world over the 10-year period between January, 2014 and December, 2023. Figure 1 presents monthly trends from Google searches of health and climate change across the five different languages and using the topic-category approach, from which we observed an upward trend in online information-seeking over this time period. The figure shows a particularly sharp increase in global engagement



**Figure 2:** Global monthly search rates of “Climate change”, “health”, and “climate change health”, and all three search terms together in English, 2014–23

Search rates of the three search terms individually and relative to each other.

with health and climate change since 2020. Focusing on English-language searches, we found that the average monthly search ratio between January, 2014 and December, 2019 was 7·4, while between January, 2020 and December, 2023 the average monthly search ratio increased to 12·4. Hence, the first insight from this analysis is that public engagement with health and climate change around the world has increased over the last 10 years, particularly since 2020.

Figure 1 shows that there was a peak in Google searches of health and climate change in April, 2022 (although Spanish-language searches feature a slightly earlier peak in October, 2021). The April, 2022 peak coincides with a large spike in searches of climate change more broadly (ie, searches of “climate change” without health) that is seen in figure 2, which presents trends in engagement with health, climate change, and health and climate change. The figure shows that relative search rates for “climate change health” are considerably lower than for either “health” or “climate change” on their own, with Google searches for “health” over this period ten-fold higher than searches for “climate change” or “climate change health”. Examining the related queries and topics in April, 2022 indicates that a combination of factors is associated with the surge in Google searches of climate change (appendix pp 12–13). In particular, there was considerable global interest in Earth Day on April 22, 2022 largely driven by Google choosing to mark Earth Day on its homepages with stark images of the climate crisis (ie, using Google Doodle, which features in the related

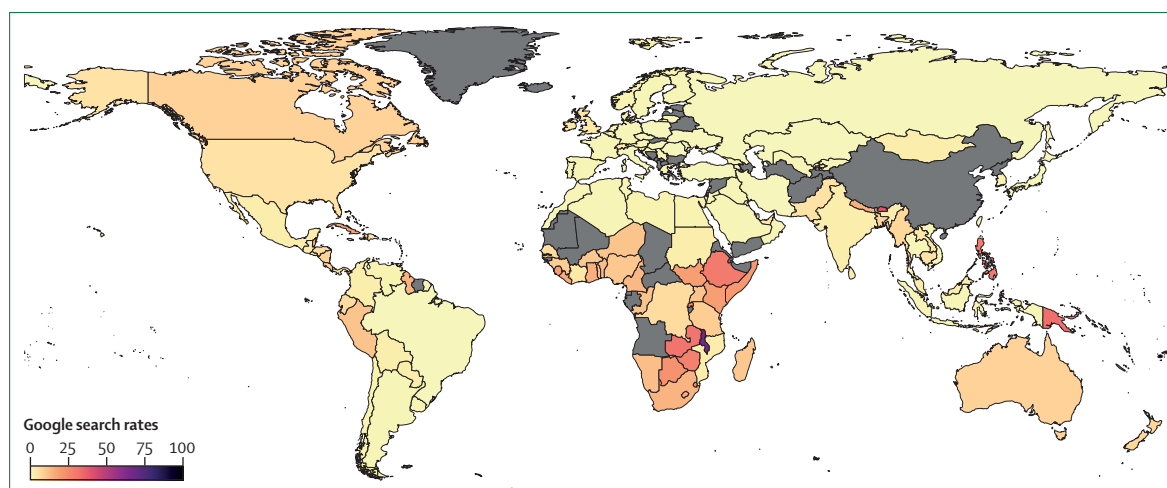
queries).<sup>38</sup> Another factor was the worldwide protest by scientists against the lack of climate action following the release of an Intergovernmental Panel on Climate Change report on the need for urgent cuts in greenhouse gas emissions by 2025 to prevent catastrophic climate impacts (with “scientist protest” featuring in the related queries during this period).<sup>39</sup> This spike in Google searches of climate change in April, 2022, in turn, is associated with increased searches of health and climate change.

While the peaks in public engagement with health and climate change closely mirror the highest engagement with climate change more broadly, the wider trends in Google searches of health and climate change differ from trends in searches of climate change. For instance, the general increase in engagement with health and climate change over the 10-year period was not observed for engagement with climate change (figure 2). Indeed, the related queries and topics for health and climate change over the 10-year period include terms specifically related to health and climate change, and broader climate change-related searches (eg, “Lancet countdown on health and climate change”, “planetary health”, and “Paris agreement”). As such, our second insight was that while peaks in global public engagement are largely the result of spikes in public engagement with climate change more broadly, the trend of increasing Google searches of health and climate change is specifically due to growing engagement with the intersection of “climate change health”, and largely distinct from engagement with either “climate change” or “health” separately.

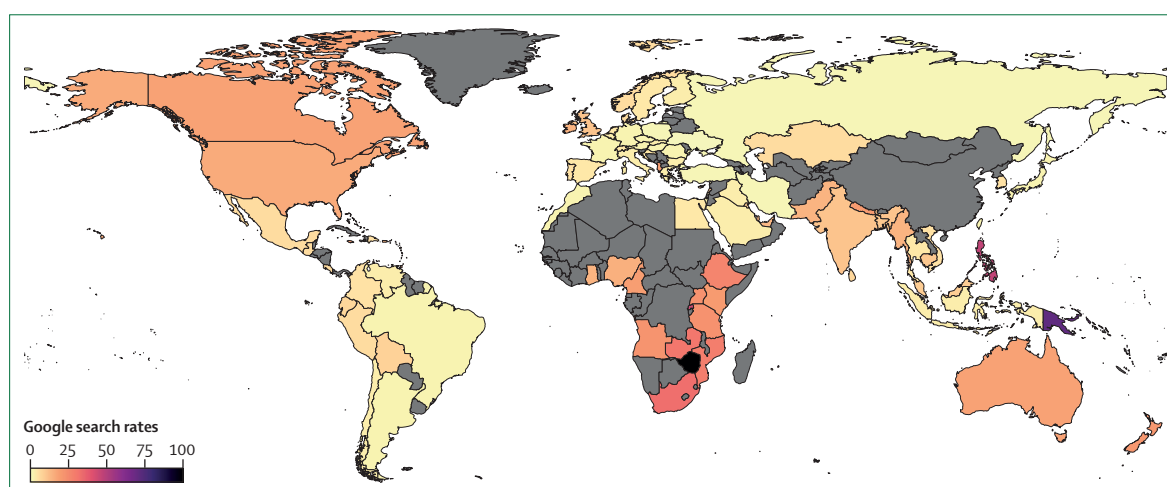
### Engagement across countries

We next considered differences in the levels of public engagement around the world, by examining which countries have the highest proportions of Google searches of health and climate change. Figures 3 and 4 show maps of global search rates between 2014 and 2023 based on Google searches of “health” and “climate change”. Figure 3 is based on a search of the terms in three languages (ie, English, Spanish, and French), while figure 4 is based on the topic–category approach with the combination of the climate change topic and public health category described earlier (we provide additional maps using different languages in the appendix pp 8–12). To more clearly present which countries have the highest proportions of Google searches of health and climate change—particularly because they include small island developing states (SIDS) that are not visible on the maps—we also include a panel listing the 20 countries with the highest levels of engagement based on Google Trends data using these two approaches (panel).

The maps in figures 3 and 4, together with the country lists in the panel, show that the highest rates of Google searches of “health and climate” change occur in LMICs rather than HICs. This finding does not imply that individuals in HICs do not perform Google searches of



**Figure 3:** Map of global search rates of “climate change health” between 2014 and 2023 in English, French, and Spanish  
Darker colours indicate a higher relative rate of searching on Google compared with other countries. Grey indicates no available data.



**Figure 4:** Map of global search rates of “climate change health” between 2014 and 2023 based on the topic-category approach  
Darker colours indicate a higher relative rate of searching on Google compared with other countries. Grey indicates no available data.

health and climate change (see figure 2), but the highest engagement is in LMICs—particularly in SIDS and African countries. One concern might be that these differences are primarily due to varying internet penetration rates across countries, whereby engagement in LMICs reflects online information-seeking from higher educated or affluent sectors of society. However, around a third of the LMICs listed in the panel have internet penetration rates higher than 70% (appendix pp 14–15), which would suggest that this finding is not simply the result of differences in internet penetration rates. Furthermore, in the appendix (p 8), we present similar maps for Google searches of “climate change” and of “health” separately and find that this difference in Google search interest between HICs and LMICs cannot be observed, and hence this global difference in engagement is specifically related to the health dimensions of climate change. As such, our third insight

is that public engagement with health and climate change is highest in lower-income countries that have contributed least to global greenhouse gas emissions, but whose populations are most susceptible to, and often most impacted by, the health effects of climate change; with lower levels of public engagement in HICs that have contributed the most to the global greenhouse gas emissions that have fuelled the climate crisis.

#### Comparisons with other search terms

A limitation of using Google Trends data to examine engagement with health and climate change is that it is based on transformed and scaled Google search data, and not data on the actual number of searches. Hence, while figure 1 shows increasing Google searches of health and climate change, it does not tell us whether this increase represents high or low levels of engagement. To partially address this, we compared engagement with



**Panel: List of 20 countries with the highest Google search rates of health and climate between 2014 and 2023 based on searches in English, Spanish, and French and the topic-category approach**

**Highest engagement using “climate change health” search term (English, Spanish, and French)**

Samoa, Fiji, Malawi, Bhutan, The Gambia, Papua New Guinea, Zambia, Philippines, Ethiopia, Zimbabwe, Rwanda, Eswatini, Botswana, Somalia, Sierra Leone, South Sudan, Kenya, Cuba, Liberia, and Belize.

**Highest engagement using “climate change” topic and “public health” category**

Zimbabwe, Papua New Guinea, Philippines, Rwanda, Mauritius, South Africa, Zambia, Mozambique, Ethiopia, Tanzania, Uganda, Angola, New Zealand, Nepal, Kenya, Trinidad and Tobago, Cameroon, Singapore, Jamaica, and Albania.

health and climate change to other climate change-related issues and other health-related themes. We did this by comparing search rates for health and climate change with other search terms. This comparison was also based on the transformed and scaled data, whereby the highest number of Google searches for either search term is assigned a value of 100, and all other datapoints for both search terms are expressed as a proportion of this peak.

Figure 5 presents pairwise comparisons based on the health and climate change searches with a range of other relevant search terms at the global-month level over the 10-year period, focusing on English-language searches. This analysis allowed us to compare search rates of health and climate change with searches of other climate change-related and climate change and health-related terms. The comparison search terms included other climate change impacts and framings (eg, “climate change impacts”, “climate change security”, “climate change economic”, and “climate change food”) and other search terms related to health and climate change (eg, “air pollution health” and “heat health”). We present comparisons with additional search terms in the appendix (p 7) based on other search terms related to climate change and health (eg, “climate change disease”, “climate change migration”, “climate change hoax”, “planetary health”, “global warming health”, etc) and variations of search terms used here (eg, “climate change economy” rather than “climate change economic”).

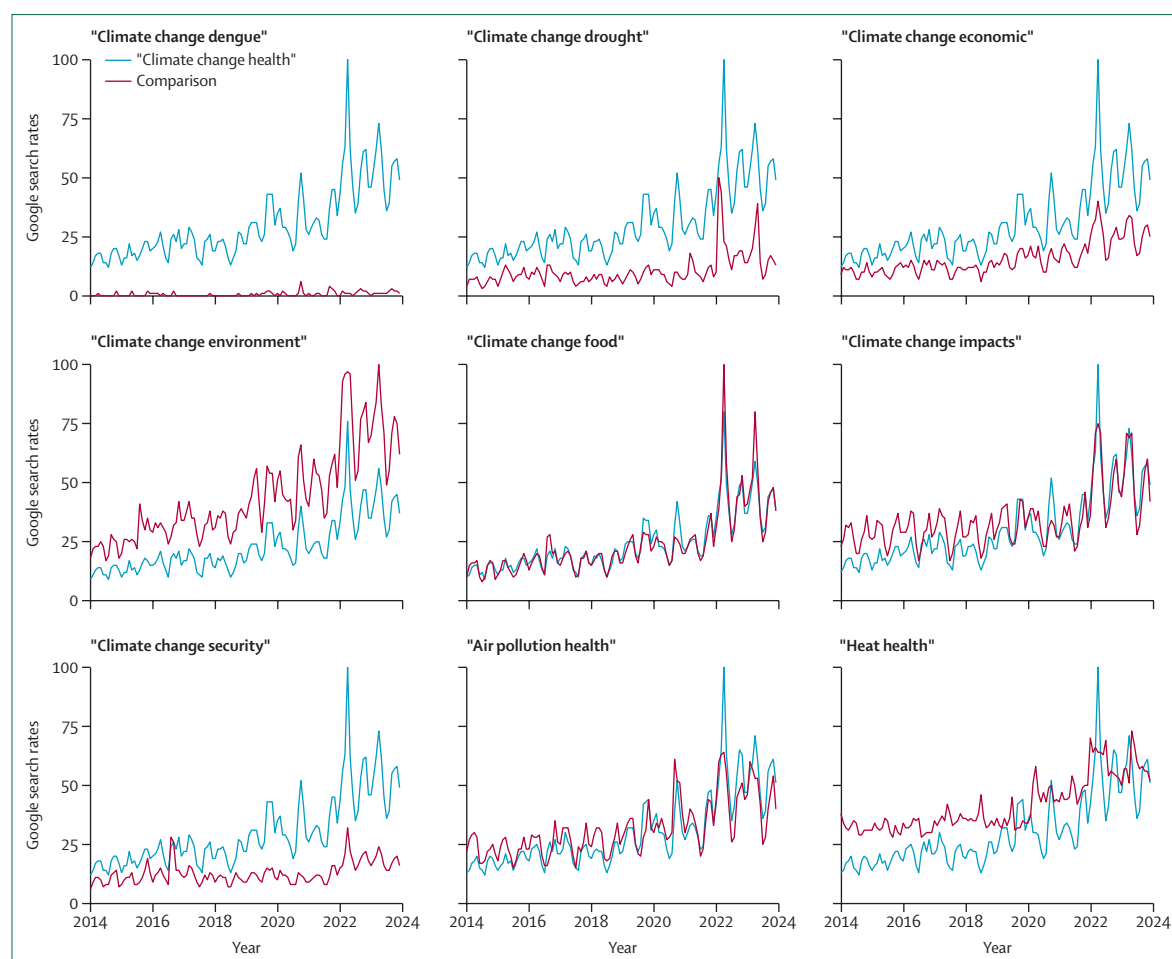
Figure 5 shows a greater engagement with “climate change health” than a range of other relevant climate change impacts and framings. For example, there was a higher proportion of Google searches of “climate change health” than “climate change economic” and “climate change economy” (appendix p 7), “climate change security”, and “climate change drought”.<sup>40</sup> We found that

engagement with “climate change health” via Google searches is similar to “climate change food” and “climate change impacts”. Engagement with “climate change impacts” was higher than “climate change health” until 2019, after which engagement for the two terms was similar. Online searches of “climate change health” were consistently lower than “climate change environment” during this period, although there has been some convergence in engagement with both search terms since 2019. We find considerably higher engagement with climate change health than with linking climate change to specific climate-sensitive diseases, such as dengue or malaria (appendix p 7). However, it is worth noting that engagement with climate change and health is much lower than for engagement with these diseases more generally (ie, not linking the diseases to climate change). Turning to health and climate change-related terms, we found that engagement with “air pollution health” is similar to “climate change health”, and while there was considerably higher engagement with “heat health” than “climate change health” previously, since 2019, the engagement with the two search terms is at similar levels. The results in figure 5, together with the trends shown in figure 2, led to a fourth insight, which is that while public engagement with “climate change health” is lower than for separate searches for “health” and “climate change”, there is greater public engagement with “climate change health” than for other climate-related frames and impacts, such as the economy, security, and drought.

## Discussion and conclusion

Greater public awareness of the relationship between climate change and human health has the potential to considerably strengthen public support for climate action. Yet, we know little about people’s engagement with health and climate change around the world, especially outside of a small number of HICs.<sup>11</sup> We address this crucial gap using individuals’ information-seeking behaviour on the Google search engine. Our analysis of Google Trends data provides several important insights regarding public engagement with health and climate change around the world.

First, we find that over the past decade—and particularly since 2020—there has been growing public engagement via Google Searches with health and climate change around the world. The existing literature on public engagement with health and climate has produced mixed findings on trends in public awareness of the health–climate change relationship. While some studies, focusing on the USA, find evidence of increasing public engagement,<sup>16</sup> other studies find little evidence to support this view.<sup>19,40,41</sup> Our global analysis provides strong evidence that there is growing public engagement with health and climate change. This evidence is also consistent with findings that indicate rising engagement with the health–climate change relationship among



**Figure 5: Pairwise comparisons of relative search rates in English, 2014–23**

Comparison of global search rates (ie, transformed volume of search volumes as a proportion of all Google searches globally).

other societal actors, such as national governments and the media.<sup>41–43</sup>

Second, our analysis indicates that although peaks in global engagement are largely driven by spikes in engagement with climate change more generally, the overall trend of increasing engagement with “climate change health” that we observe over the 10-year period is distinct from trends in engagement with either “climate change” or “health” separately. This finding is important because it suggests that growing public engagement with health and climate change is not simply a consequence of greater public interest in climate change more broadly, but that there is increasing engagement with the human health–climate change relationship specifically. In turn, this finding suggests that the global narrative about climate change could be turning to a more human and personal dimension, with the potential, over time, to engage more people with climate change-related issues. Furthermore, the finding suggests that efforts to increase public awareness of the health dimensions of climate change have had some effect.

Third, and particularly importantly, we find that engagement with health and climate change via Google searches is highest in LMICs, especially in SIDS and African countries. Hence, engagement is highest among the regions most affected by the impacts of climate change, and in those that have contributed least to the problem. In contrast, we find lower levels of public engagement with health and climate change in HICs and high-emitting countries. The few surveys of public perceptions of health and climate change that have been conducted in different LMICs suggest somewhat high levels of public awareness of health risks.<sup>44,45</sup> Our analysis shows high public engagement across a wide range of LMICs over a 10-year period and that engagement with health and climate change, based on proportions of Google searches, is higher than in HICs. In other words, public engagement is highest in places where the health impacts of the climate crisis are most acutely felt. This finding is consistent with governments of LMICs being more engaged with the health dimensions of climate change in global governance arenas than governments of

HICs and suggests a potential relationship between government and public engagement with health and climate change that future research could consider more closely. This finding also highlights stark global inequalities in information production on health and climate change and information-seeking behaviour, whereby existing studies indicate that in scientific research and global reports insufficient attention is given to the countries most affected by and susceptible to the climate crisis; while it is the public in these countries that most seek information about the connections between climate change and human health.<sup>41,46,47</sup>

Lastly, we also considered how engagement with health and climate change compares to engagement with climate change-related and other health-related issues. Predictably, our results indicate that engagement with “climate change health” is considerably lower than engagement with “health” and “climate change” separately. Yet, our analysis finds that engagement with health and climate change is higher than with other climate change-associated issues, such as those related to the economy, conflict, and security. This finding is consistent with those that suggest a health frame can increase exposure to climate change news among some groups in the USA more than other climate change frames.<sup>48</sup> Again, our results suggest growing engagement with the relationship between health and climate change among the global public, and the increasing salience of a health framing of climate change.

Taken together, our findings provide important new insights on public engagement with health and climate change around the world. It is, of course, important to recognise the limitations of the use of Google Trends data to measure public engagement, as we have discussed. Google Trends data capture a specific dimension of public engagement, namely online information-seeking behaviour. Furthermore, the unavailability of raw search volume data prevents a more complete understanding of online information-seeking about climate change and health. It is also worth noting the potential effect of emerging artificial intelligence tools on Google search behaviour and, consequently, on Google Trends data. As artificial intelligence assistants become more sophisticated and widely adopted, they might fundamentally alter how individuals seek information online. Such tools could reduce direct Google searches by providing immediate answers, potentially leading to lower overall search volumes, or conversely, they might drive more complex, nuanced searches that artificial intelligence tools cannot fully address, and could vary across regions or groups. These evolving dynamics underscore the need for cautious interpretations of Google Trends data in future research and emphasise the importance of considering multiple data sources—including surveys, social media, and interviews—to comprehensively assess public engagement with health and climate change issues. Yet, the

global coverage and availability over time of Google Trends data means that it can provide us with an indicator of public engagement with health and climate change around the world that can be monitored over time and can be used to assess obstacles and drivers of such engagement in specific contexts. This resource is particularly important given that public engagement is crucial for strengthening climate commitments around the world.

#### Contributors

All authors contributed to the conceptualisation and design of the study. DKP conducted the data collection, analysis, and visualisation. SJ contributed to the data analysis. All authors contributed to the data interpretation. ND wrote the first draft of the manuscript and all authors contributed to the final manuscript and approved it for publication.

#### Declaration of interests

ND received institutional funding from the EU’s Horizon Europe programme at the University of Birmingham. DKP is a postdoctoral researcher at the University of Birmingham funded by the EU’s Horizon Europe programme. MR is employed by University College London and is funded by a Wellcome Trust grant and the EU’s Horizon Europe programme. This funding had no role in the development of the article nor the decision to submit for publication. SJ declares no competing interests.

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